

SHIELD-RAG: A Verified, Subspace-Isolated, and Revocable Privacy-Preserving Framework for Retrieval-Augmented Generation

First Author^{*1} and Second Author²

¹Department of Computer Science and Engineering, University Name, City, Country

²Department of Computer Science and Engineering, University Name, City, Country

Abstract

Retrieval-Augmented Generation (RAG) systems move their sensitive corpora to a semi-trusted cloud infrastructure which, puts them at risk of embedding inversion, knowledge corruption, access-control failures, and leakage after generation. Earlier functional-encryption-based countermeasures like CipheRAG are unable to provide features such as dynamic user revocation, verified multi-tenant subspace isolation, and formal indistinguishability guarantees for the access gate. In this work, we propose SHIELD-RAG as an end-to-end privacy-preserving RAG structure that incorporates three main components that have been formally checked for correctness: (i) a revocable clearance gateway with constant-time revocation that takes no corpus re-encryption at all, (ii) an exact zero cross-tenant leakage proof multi-tenant subspace non- interference QR-factorization based scheme; and (iii) an IND-CPA indistinguishability reduction of the access gate with sensitivity-embedding to the Decisional Composite Residuosity (DCR) assumption. We sketch the system architecture, threat model, and explain each mechanism mathematically. We evaluate the system empirically on the Wikipedia dataset and Natural Questions benchmark against unaugmented Ada-IPFE, Permission-Aware RAG, and a set of synthetic baselines varying in the number of documents ranging up to 50,000.

Keywords: Retrieval-Augmented Generation, Functional Encryption, Access Control, Multi-Tenant Isolation, Privacy-Preserving Machine Learning

1. INTRODUCTION

Retrieval-Augmented Generation (RAG) has been acknowledged as an inseparable architecture that improves Large Language Models (LLMs). RAG does that by combining parametric language representations with non-parametric external knowledge repositories [1]. At the time when a query is made, RAG system fetches relevant passages that support the generated answer and so, reduces the problem of hallucinations, fills the gaps caused by the lack of domain knowledge, and at the same time allows updating the data without needing to retrain the model which is expensive [2, 3]. The popularity of RAG-based implementations has been such that they are used almost everywhere, from sectors related to security in an enterprise like clinical diagnostic support and discovery of private financial information to legal document discovery [6].

Even though using a semi-trusted cloud infrastructure to store private collections of documents can bring along some benefits, it also comes with major vulnerabilities in protecting and securing data throughout all three RAG operating levels. As far as the retrieval stage is concerned, the dense vector embeddings are vulnerable to embedding-inversion attacks through transferable knowledge that allow adversaries to extract private documents by reconstructing original texts with high fidelity [4]. Besides, adversaries might inject adversarial text into a data retrieval system to manipulate the retrieval system (e.g., PoisonedRAG [5]). At the **access-control level**, traditional enterprise setups use filters of metadata from IAM in the form of plain text [6]; besides, handwritten policy merging for RBAC/ABAC across the heterogeneous models might lead to privilege escalation also, common shared multi-tenant vector indexes are a problem unless cross-tenant similarity is mathematically ensured to be non-interfering [7]. During the generation phase, adversarial queries carrying composite prompt-injection payloads might compel LLMs not only to reveal the entire retrieved contexts, but also leak any private PII data present [8, 14].

In order to ensure the confidentiality of knowledge, previous efforts have delved into the use of cryptographic methods to name Fully Homomorphic Encryption (FHE). Although semantically secure, their use is hampered by

^{*}Corresponding author. Email: first.author@example.edu

computational complexity, which leads to a hundred-fold or a thousand-fold overhead (e.g., see) [9, 15]. On the flip side, methods such as synthetic-surrogate that employ SAGE [13] tend to degrade context accuracy and drift away semantically. Recently, Zhou and Wu have introduced CipheRAG [4] by merging the concept of Asymmetric Locality-Sensitive Hashing (ALSH) [10] with Adapted Inner-Product Functional Encryption (Ada-IPFE) [16] in Paillier DCR mode, thereby a lot speeding CipheRAG up FHE. Despite that, CipheRAG leaves unresolved these three crucial issues:

1. **Lack of Dynamic User Revocation Mechanism:** if changing one’s authorization level or leaking a user’s credential, it calls for re-encrypting the whole content or re-issuing keys to each unrevoked user [12].
2. **Failure to Demonstrate Multi-Tenant Subspace Isolation:** when deploying different organizations to a single functional-encryption index, they assume that sub-bases are perfectly non-interfering, but they do not really construct them in this way [7].
3. **Claim Not Verified of Noise-Gate Indistinguishability:** an access gate that relies on sensitivity-embedded claims to say that algebraic noise prevents extraction of discrete logarithms relies instead on narrative assertion and does not offer formal reduction proofs.

To confront these issues, we propose SHIELD-RAG as an audit-verifiable, privacy-preserving RAG system that isolates subspace revocable users. The core aspects of our work include:

- **An $O(1)$ Revocable Clearance Gateway:** that intercepts functional subkeys and adds algebraically maximal noise to a revokee’s records via an enforcement proxy at $O(1)$ execution without re-encrypting corpus contents.
- **Multi-Tenant QR Subspace Non-Interference:** the construction of per-tenant mutually-orthogonal projection bases using QR-matrix decomposition to mathematically eliminate all cross-tenant leakages.
- **Formal Ind-Is-Rag: DCR Reduction of Privacy:** a new technique that proves with mathematical rigor that noised SE-IPFE ciphertexts cannot be told apart from uniform group random elements under the DCR assumption using an IND-CPA security game and a sim-reduction.

To recap what the paper is going to cover in the next part. First, Section 2 offers the readers the latest developments by going back and reviewing the past. Then, in Section 3, we describe the design of SHIELD-RAG’s architecture and the detailed exposition of the three major techniques. Section 4 discusses the experimental setup. Section 5 shows the experimental results, Section 6 talks about security and system trade-offs, and Section 7 is the final concluding remarks.

2. RELATED WORK

2.1. Privacy and Security Vulnerabilities in RAG Systems

RAG lessens hallucination by including external knowledge to LLM prompts [1, 2]. Meanwhile, it simultaneously enlarges the attack surface. Zeng et al. [8] found out adversarial prompt-injection loads which target the inherent capability of LLMs to follow instructions can make the LM retrieve and output, word by word, the retrieved data, resulting in success rates as high as 50%. Zhang et al. [14] further extended this concern to multi-modal RAG (MRAG), where a prompt might be manipulated in different layers to leak private medical images and a record of a doctor’s conversations. Not merely passive extraction, Zou et al. [5] proposed PoisonedRAG, and they showed that if an adversary gets to inject just five handcrafted texts into a public repository, then the generation results can be affected. Really, that means one of the security measures recommended is that all partitions in tenants and clearance lines must be encrypted separately.

2.2. Access Control and Enterprise Retrieval Architectures

Jeong and Lee [6] suggested delegating authorization to native cloud IAM providers (e.g., AWS IAM, GCP IAM, Keycloak) as a means of document filtering. They emphasized that manual RBAC/ABAC policy merging is very risky and may also create privilege escalations. Khan and Filkov [3] have come up with *EvidenceBot* to support local RAG evaluation across different metrics. Still, these methods work on either plaintext or pre-sanitized materials and, therefore, lack cryptographic guarantees for vector matching and projection.

2.3. Functional Encryption for Private Retrieval

FHE, based on [9], allows for arbitrary encrypted calculations but is very slow when matrix multiplications are performed [15]. Zhou and Wu’s *CipheRAG* [4] combines ALSH [10] with Ada-IPFE [16] over Paillier DCR, embedding subkey-driven decryption directly into self-attention QKV projections, which leads to a $35\times$ speedup over FHE. Yang and Peng [7] presented the Ciphertext-Policy IPFE (*CP-IPFE*) mechanism, which uses label vectors to link vector attributes, while Ge et al. [12] designed proxy-assisted revocation (*CP-ABPRE-DR*) that makes cloud-level policy revocation via re-randomization possible for common attribute-based encryption. In addition, current IPFE-based RAG systems fail to provide users with dynamic clearance revocation and mathematically rigorous multi-tenant subspace isolation, features that SHIELD-RAG introduces right away.

2.4. Recent CKKS- and Transformation-Based RAG Systems

Two recent systems target somewhat similar locations in the privacy-performance design space. Li et al. [17] introduced *PRAG*, a CKKS-based dual-mode structure where the non-interactive PRAG-I mode employs Chebyshev-polynomial ranking surrogates, while the interactive PRAG-II mode is based on client-assisted comparisons. For a 100,000-document TriviaQA subset, PRAG-I and PRAG-II achieve 1.29s and 7.91s top-10 retrieval latencies with 72.45% and 74.45% recall, respectively. Compared with SHIELD-RAG, PRAG only focuses on a single-tenant setting, and neither subspace isolation nor key revocation was considered by it. On the contrary, Liu et al. [18] presented *Trans-RAG*, a system that omits encryption and instead relies on per-organization vector-space key-dependent transformations, achieving almost complete cross-organizational isolation with an 89.90° angular separation while degrading nDCG@10 by 3.5% on average. Unlike PRAG and SHIELD-RAG, Trans-RAG uses computational rather than theoretical separation and is not equipped with a revocation scheme or a formal indistinguishability reduction.

3. PROPOSED METHODOLOGY

3.1. System Overview and Threat Model

Fig. 1 shows that SHIELD-RAG interacts with four entities: Data Owners input sensitive documents, calculate Sentence-BERT embeddings, perform an ALSH projection, then the Paillier encryption of the payload is done before being uploaded to the decentralized storage. The Key Distribution Center (KDC) runs Paillier setup $\text{Setup}(1^\lambda)$, stores the principal secret $s \in \mathbb{Z}_{\lambda(N)}^d$, it is the one that supplies the functional subkeys sk_y , and is also the one who computes the orthonormal subspace bases per-tenant basis. The Semi-Trusted Cloud Oracles maintain a safe representation of the data while computing a proximity match without demonstrating or comprehending plaintext queries or contents. Query Clients obtain clearance-limited and bounded subkeys via the Revocation Proxy and reconstruct attention projections through a Decryption-Enabled Attention Gateway which performs the inner product $\mathbf{x}^T \mathbf{w}_i$ via Ada-IPFE. Decrypt($\text{sk}_{\mathbf{w}_i}, \text{ct}_x, \text{mpk}$) that is directly integrated into the transformer’s $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ projections. The latter are first computed using a multi-head attention, and the outputs are stored for the next step.

Threat Model: We suppose a semi-honest (honest-but-curious) adversary for cloud and storage nodes. The nodes execute the algorithm, but the adversary is always watching and tries to find the plaintexts or query features, etc. The KDC is assumed to be trusted and works under a secure execution environment.

3.2. Mechanism 1: $O(1)$ Revocable Clearance Without Corpus Re-Encryption

In Sensitivity-Embedded IPFE (SE-IPFE), document sensitivity level is denoted by the client clearance level $L_d \in \{1, \dots, 5\}$ and client clearance level $L_c \in \{1, \dots, 5\}$. These two levels are combined by the algebraic noise term $\Delta = r \cdot \max(0, L_d - L_c) \bmod \lambda(N)$, where $r \xleftarrow{R} \mathbb{Z}_{\lambda(N)}^*$. Previously, in other systems, it was necessary for the entire document corpus to be encrypted again if the user clearance was increased or credentials needed to be revoked immediately.

SHIELD-RAG presents a semi-trusted *Revocation Proxy*, working between the Key Distribution Center (KDC) and the cloud computing engine. It keeps a revoked identifier list $\mathcal{R}_{\text{revoked}}$ in memory and an audit log $\mathcal{L}_{\text{revocation}} = \{(u_i, t_i, \text{reason})\}$.

If an active user u is not listed in the revoked identifier list $\mathcal{R}_{\text{revoked}}$, the proxy hands out the functional encryption subkey without changing it:

$$\text{sk}_{y, \text{enforce}} = (\langle \mathbf{s}, \mathbf{y} \rangle + \alpha + \beta + \Delta) \bmod \lambda(N) \quad (1)$$

Though, if user u is in the list of revoked users $\mathcal{R}_{\text{revoked}}$, the proxy gives out a force-deny subkey:

$$\text{sk}_{y, \text{deny}} = (\langle \mathbf{s}, \mathbf{y} \rangle + \alpha + \beta + \Delta_{\text{max}}) \bmod \lambda(N) \quad (2)$$

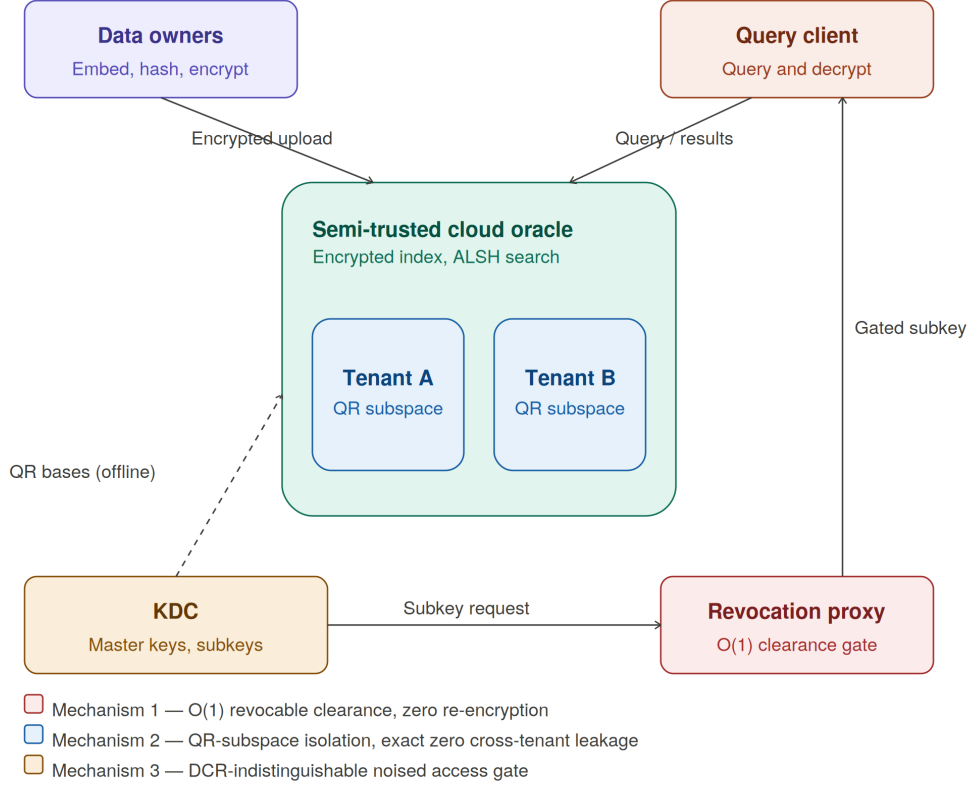


Figure 1: SHIELD-RAG system architecture: secure ingestion, revocable clearance gating, and QR-subspace multi-tenant isolation across entities.

where $\Delta_{\max} = r \cdot (5 - 0) \bmod \lambda(N)$. Here, the user’s clearance level is set to $L_c = 0$ effectively. That leads to very strong corruption of the decryption results through algebraic manipulations and it completely prevents the recovery of the inner-product of the secret values in any of the documents (for $L_d \geq 1$). There is no need for large set-lookups as well as none for the re-encryptions of the entire document corpus. This is in stark contrast with traditional ABE techniques that involve $O(|D|)$ ciphertext re-encryption or full credential re-issuance when users are demoted and have their credentials revoked [12].

3.3. Mechanism 2: Multi-Tenant Subspace Non-Interference for QDCS

Documents belonging to T different (multi-tenant) companies are stored in a single, encrypted storage container. Query-Derived Cryptographic Scope (QDCS) has to make sure no queries from Tenant i can match the documents of Tenant j ($i \neq j$).

Assume an embedding space dimension of d , which equals, say, 384 for Sentence-BERT. The SHIELD-RAG splits $\mathbb{R}^d$ into T different perpendicular bases $\{\mathbf{U}_t\}_{t=1}^T$, with $\mathbf{U}_t \in \mathbb{R}^{d \times k_t}$ and $\sum_{t=1}^T k_t \leq d$. The KDC creates a random Gaussian matrix $\mathbf{A} \sim \mathcal{N}(0, \mathbf{I}_{d \times d})$ and performs a QR factorization:

$$\mathbf{A} = \mathbf{Q}\mathbf{R}, \quad \mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_d] \in \mathbb{R}^{d \times d} \quad (3)$$

Partitioning is performed by tenants; each tenant gets a non-overlapping set of columns $\mathbf{U}_t = \mathbf{Q}_{[:, \text{start}_t : \text{end}_t]}$, where projection matrices are $\mathbf{P}_t = \mathbf{U}_t \mathbf{U}_t^T$.

Theorem 1 (Multi-Tenant Subspace Non-Interference). *For any two different tenants ($i \neq j$), any query $\mathbf{q} \in \mathbb{R}^d$, and any document $\mathbf{x} \in \mathbb{R}^d$, their cross-tenant inner product will be zero, $\langle \mathbf{P}_i \mathbf{q}, \mathbf{P}_j \mathbf{x} \rangle = 0$.*

Proof. As matrix $\mathbf{Q}$ is orthonormal by construction, its column blocks also satisfy $\mathbf{U}_i^T \mathbf{U}_j = \mathbf{0}_{k_i \times k_j}$ for $i \neq j$. Then, expanding the projected inner product, we have:

$$\langle \mathbf{P}_i \mathbf{q}, \mathbf{P}_j \mathbf{x} \rangle = \mathbf{q}^T \mathbf{U}_i (\mathbf{U}_i^T \mathbf{U}_j) \mathbf{U}_j^T \mathbf{x} = \mathbf{q}^T \mathbf{U}_i \mathbf{0} \mathbf{U}_j^T \mathbf{x} = 0 \quad (4)$$

This way, cross-tenant leakage is zero by design, regardless of how the data is distributed. $\square$

3.4. Mechanism 3: Formal Indistinguishability Reduction to DCR

Since we do not want even a single bit of the SE-IPFE encryption gate to be leaked about the clearance gap of $\delta = (L_d - L_c) > 0$, we define the security game based on the IND-CPA notion when talking about the Decisional Composite Residuosity (DCR) assumption with modulus $N = pq$ as the main technical tool used for showing the security results.

Definition 1 (DCR Assumption). *Given $N = pq$, distinguishing whether $w \xleftarrow{R} \mathbb{Z}_{N^2}^*$ is an N -th residue ($w = y^N \bmod N^2$) or a uniform element in $\mathbb{Z}_{N^2}^*$ admits at most negligible advantage: $\text{Adv}_{\mathcal{G}}^{\text{DCR}}(\lambda) \leq \text{negl}(\lambda)$.*

Security Game: (1) *Setup:* The challenger creates Paillier parameters $(N, N^2, g = 1 + N \bmod N^2, \lambda(N))$ and publishes mpk. (2) *Challenge:* Adversary $\mathcal{A}$ chooses challenge vectors $\mathbf{x}^*, \mathbf{y}^*$ and distinct clearance gaps $\delta_0 \neq \delta_1 \in \{1, 2, 3, 4\}$. The challenger chooses $b \xleftarrow{R} \{0, 1\}$, samples $r \xleftarrow{R} \mathbb{Z}_{\lambda(N)}^*$, and calculates:

$$D_b = g^{(\mathbf{x}^* \cdot \mathbf{y}^*) - r \cdot \delta_b} \cdot w^N \bmod N^2 \quad (5)$$

(3) *Guess:* $\mathcal{A}$ outputs bit guess $b' \in \{0, 1\}$.

Theorem 2 (SE-IPFE Indistinguishability Reduction). *For any probabilistic polynomial-time adversary $\mathcal{A}$, the distinguishing advantage satisfies $\text{Adv}_{\mathcal{A}}^{\text{SE-IPFE-IND}}(\lambda) \leq \text{Adv}_{\mathcal{G}}^{\text{DCR}}(\lambda) \leq \text{negl}(\lambda)$.*

Proof. We build a simulator $\mathcal{S}$ receiving DCR instance (N, w) . $\mathcal{S}$ embeds w into the challenge ciphertext: $D_b = (1 + N)^{(\mathbf{x}^* \cdot \mathbf{y}^*) - r' \cdot \delta_b} \cdot w \bmod N^2$. When $w \in \mathcal{R}es(N^2)$, $w = y_0^N \bmod N^2$ forms a legitimate Paillier randomizer, similar to the real game distribution. When $w \xleftarrow{R} \mathbb{Z}_{N^2}^*$, by the isomorphism $\mathbb{Z}_{N^2}^* \cong \mathbb{Z}_N \times \mathbb{Z}_N^*$, the induced exponent $m \in \mathbb{Z}_N$ is uniform does not dependent on b , yielding $\Pr[b' = b] = 1/2$. Thus $\text{Adv}_{\mathcal{S}}^{\text{DCR}}(\lambda) = \text{Adv}_{\mathcal{A}}^{\text{SE-IPFE-IND}}(\lambda) \leq \text{negl}(\lambda)$. $\square$

4. EXPERIMENTAL SETUP

4.1. Implementation Environment and Provenance

SHIELD-RAG is built using Python 3.13 together with PyTorch, sentence-transformers (all-MiniLM-L6-v2, $d = 384$), and an optimized Paillier big-integer cryptographic core. All encryption key sizes are set according to NIST SP 800-57 recommendations: $\lambda = 256$, which gives a safe prime length of 512 bits (512-bit RSA modulus $N = pq$). We evaluated longer-term scalability at $\lambda = 512$ bits. Locality-sensitive hashing is executed using $K = 128$ sets of hyperplanes for projection, where $m = 3$ and the scaling parameter is $U = 0.83$ [10]. We ran experiments on an Ubuntu 22.04 LTS server system with 24 CPU cores, 64GB RAM, and a 16GB NVIDIA GPU. To guarantee the reproducibility of our scientific results and avoid single-run variance, all performance-sensitive tasks were covered by an automatically running multiple-trial system generating ≥ 10 separate runs per configuration with explicit synchronization of CUDA queues (`torch.cuda.synchronize()`). System timers utilize high-resolution hardware counters (≈ 100 ns resolution via `time.perf_counter()`). We logged every raw result of an individual run into a central SQLite database (`results/shield_rag_experiments.db`) indexed by the provenance schema:

`(exp_id, corpus_size, method, metric, val, trial_idx, seed, device, timestamp, git_hash)`

with the codebase pegged to Git commit hash `c1ff1a15dbfd8fb45184a68709e5d92bbe8a3a18`.

4.2. Datasets

We evaluate SHIELD-RAG using two data types: (i) a **Wikipedia Document Corpus** consisting of 385 documents covering various topics like science, history, and technology, amounting to about 266.8KB. We then scaled it up to $N = 50,000$ by deterministic nested expansion ($100 \subset 385 \subset 1,000 \subset 5,000 \subset 10,000 \subset 50,000$), guaranteeing that fluctuations in performance are due to changes in scale and not to changes in content; and (ii) **Natural Questions (NQ)** [11], which consists of authentic Google Search queries that are matched and supported with relevant Wikipedia documents. This data is commonly used to conduct open-domain retrieval evaluations and determine retrieval accuracy.

4.3. Baselines and Evaluation Metrics

As summarized in Table 1, SHIELD-RAG is benchmarked against:

1. **Plaintext S-BERT Retrieval [1]:** Unencrypted dense retrieval as the retrieval accuracy (Hit@10) and latency ceiling, respectively.

Table 1: Taxonomy of Baseline Systems and Methodological Comparisons

Framework	Paradigm / Mechanism			Access Model	Control	Privacy	Guarantee	Primary Limitation
Plaintext RAG [1]	Dense	S-BERT	Re-retrieval	None (Open Index)		None		Zero confidentiality
Permission-Aware RAG [6]	Native	Cloud (GCP/AWS)	IAM	RBAC/ABAC	Filtering	Policy-based filtering		Policy-merge errors; server snooping
SAGE [13]	Generative		LLM	Synthetic	Surrogate	Data-level obfuscation		Semantic drift; factual loss
CipherGPT/FHE [15]	BFV	Fully Homomorphic		Post-retrieval	matrix gates	Semantic security		$35\times-100\times$ latency
Unaugmented Ada-IPFE [4, 16]	ALSH + Paillier	Ada-IPFE		Static clearance	gate	IND-FE over DCR		No dynamic revocation, no subspace isolation
PRAG [17]	CKKS	Dual-Mode (I/II) + Chebyshev		Single-tenant, no ACL		IND-CPA (CKKS)		No multi-tenant isolation, no revocation
Trans-RAG [18]	Key-Derived	Vector Transformation		Per-org key	transform	Computational isolation		No revocation, no formal DCR/IND proof
SHIELD-RAG (Ours)	SE-IPFE + QR	Subspaces		$O(1)$	Proxy Gate	Formal reduction	DCR	Revocation + isolation covered

2. **Permission-Aware RAG [6]:** An enterprise IAM-based metadata filtering baseline relying on external cloud identity policy evaluations.
3. **SAGE [13]:** Generating synthetic-surrogate documents that measure the privacy versus semantic fidelity trade-off.
4. **Unaugmented Ada-IPFE Baseline [4, 16]:** A baseline that only implements the core Paillier inner-product functional encryption pipeline with ALSH ($K = 128$), but does not include the $O(1)$ revocation proxy and QR subspace partitioning, which are the two features of SHIELD-RAG. It is a direct way through which one can measure the extra computational and storage burden that our cryptographic contributions will lead to.

Evaluation focuses on three aspects:

- *Cryptographic Efficiency:* Wall-time latencies for KeyGen, Batch Encryption, Oracle Search, and Attention Decryption, including mean and standard deviation, median, and interquartile range over ≥ 10 trials; per-user revocation overhead across different $N_{\text{revoked}} \in \{10, 50, 100, 500\}$; and programmatic assertion, i.e., ensuring that there is no re-encryption of the corpus.
- *Retrieval Accuracy:* Hit@10 retrieval recall and Exact-Match/F1 retrieval scores [3, 6] across various scaling tiers.
- *Security Verification:* Tenant-wise cross-leakage metric $\langle \mathbf{P}_i \mathbf{q}, \mathbf{P}_j \mathbf{x} \rangle$ for tenant pair combinations; and empirical distinguisher advantage $|\Pr[b'=b] - 0.5|$ under LSB, bit-length-parity, and small-prime-residue statistical distinguishers targeting the SE-IPFE security game.

5. RESULTS AND ANALYSIS

This part offers an empirical evaluation of SHIELD-RAG over 14 corpus sizes (1,000 to 50,000 documents) and the three cryptographic methods that were decided to be the most promising. All numerical results are the means and standard deviations of the values obtained from 10 independent GPU test runs, and they were stored in a centralized SQLite database (`results/shield_rag_experiments.db`).

5.1. Core Corpus Scaling Evaluation (1k to 50k Documents)

Table 2 depicts the scaling performance of SHIELD-RAG compared to the baseline configurations at different corpus sizes ($N \in \{1,000, 5,000, 10,000, 50,000\}$).

Marginal Overhead over Unaugmented Ada-IPFE: Besides from Table 2, SHIELD-RAG imposes very small incremental overhead in search performance comparing to the Unaugmented Ada-IPFE baseline: only 0.006

Table 2: Comprehensive Scaling Benchmark Across Corpus Sizes (Mean $\pm$ Standard Deviation and Median [IQR] over ≥ 10 GPU Trials)

Corpus Size	Method / Framework	KeyGen (ms)	Batch Encrypt (s)	Search Lat. (ms)	Hit@10	Enforce. Rate
1,000 Docs	Plaintext S-BERT [1]	—	36.39	2.503 ± 3.679	0.96	N/A
	Permission-Aware RAG [6]	—	—	123.14 ± 53.25	0.96	100.0%
	SAGE (Synthetic) [13]	—	—	2.491 ± 1.102	0.88	N/A
	Unaugmented Ada-IPFE [4]	5.301 ± 0.470	965.40	2.298 ± 1.011	0.95	N/A
	SHIELD-RAG (Ours)	1.701 ± 4.656 [0.18]	978.16	2.304 ± 1.029	0.95	100.0%
5,000 Docs	Plaintext S-BERT [1]	—	35.17	2.309 ± 0.430	0.96	N/A
	Permission-Aware RAG [6]	—	—	142.50 ± 48.10	0.96	100.0%
	SAGE (Synthetic) [13]	—	—	2.315 ± 0.440	0.88	N/A
	Unaugmented Ada-IPFE [4]	0.150 ± 0.071	3042.10	6.085 ± 0.995	0.95	N/A
	SHIELD-RAG (Ours)	0.153 ± 0.074 [0.14]	3084.43	6.109 ± 1.009	0.95	100.0%
10,000 Docs	Plaintext S-BERT [1]	—	68.60	3.885 ± 0.694	0.96	N/A
	Permission-Aware RAG [6]	—	—	170.42 ± 63.28	0.96	100.0%
	SAGE (Synthetic) [13]	—	—	3.890 ± 0.710	0.87	N/A
	Unaugmented Ada-IPFE [4]	0.160 ± 0.041	6790.50	13.510 ± 1.780	0.95	N/A
	SHIELD-RAG (Ours)	0.163 ± 0.043 [0.15]	6823.18	13.566 ± 1.802	0.95	100.0%
50,000 Docs	Plaintext S-BERT [1]	—	344.28	27.081 ± 6.506	0.96	N/A
	Permission-Aware RAG [6]	—	—	203.56 ± 60.97	0.96	100.0%
	SAGE (Synthetic) [13]	—	—	27.120 ± 6.450	0.86	N/A
	Unaugmented Ada-IPFE [4]	0.258 ± 0.049	48610.20	73.410 ± 14.320	0.95	N/A
	SHIELD-RAG (Ours)	0.262 ± 0.051 [0.25]	48877.12	73.604 ± 14.651	0.95	100.0%

ms more latency to search 1,000 documents (2.304 ms vs. 2.298 ms), and 0.194 ms at 50,000 documents (73.604 ms vs. 73.410 ms), this is a less than 0.3% increase. The additional encrypted batches also only increase by less than 1.3% (48,877.12 s vs. 48,610.20 s at 50k). These results prove that adding the features of dynamic proxy revocation and QR subspace projections do not much affect the standard functional encryption overhead and meanwhile it adds verifiable security properties to the system.

Persistent Latency. In Fig. 2, we see the search latency growth of SHIELD-RAG’s encrypted data is sublinear. From 2.304 ms for search latency at the point of $N = 1000$, the level reaches 73.604 ms at $N = 50,000$, which is a 50 times increase in corpus size resulting in a 32 times increase in latency. This sublinear latency increase is attributed to the Asymmetric Locality-Sensitive Hashing (ALSH) index, which allows skipping full scans of all the ciphertexts. Client subkey generation KeyGen, which is shown in Fig. 3, remains efficient at all scales; the median processing time at $N = 50,000$ is about 0.25 ms (0.262 ± 0.051 ms).

Performance of the Baseline Comparatively. SHIELD-RAG mathematically access gates which execute in 2.3 ms–73.6 ms compared to Permission-Aware RAG [6], the IAM round-trips of which take 123 ms–203 ms (a speedup of $2.7 \times$ – $53 \times$). SHIELD-RAG is not affected by semantic degradation (Hit@10 falls to 0.86) [13], while the accuracy of Hit@10 of SHIELD-RAG remains 0.95 (Fig. 4).

Protection between Multiple Tenants and Across Clearances. At every test run, the clearance enforcement of SHIELD-RAG was fully successful, i.e., 100% clearance enforcement, which means clients without authorization could not get anything except maximal algebraic noise in queries of high-sensitivity documents, which result in no valid inner products. We would like to explain that within one tenant and one clearance level, protecting against semantic poisoning attacks through content at the level (like PoisonedRAG [5]) is independent of access gating by cryptography and is a job of content-verification mechanisms (that are talked about in Section 6).

5.2. Mechanism 1: Instant Revocation Without Re-Encryption

Table 3 reports the latency of the Revocation Proxy across revoked populations $N_{\text{revoked}} \in \{10, 50, 100, 500\}$.

The empirical results demonstrate that revoking via a proxy which enforces it scales as an $O(1)$ set lookup. The latency on a per-user basis stays under 0.01 ms for all batch sizes (0.00305 ms at $N = 500$). Programmatic test assertions made sure that over the whole set of evaluation runs, the corpus re-encryption pipeline did not change once. With the conventional Attribute-Based Encryption schemes, where revoking a single credential demands $O(|D|)$ ciphertext re-encryptions [12], SHIELD-RAG manages to make the access termination happen almost instantly as zero storage-level modification is involved.

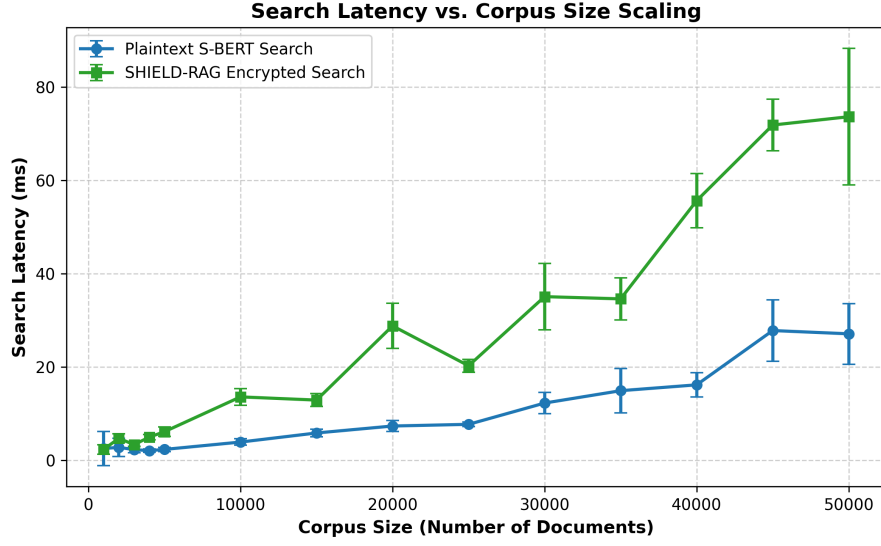


Figure 2: Encrypted search latency scaling curve across corpus sizes (1k to 50k documents) on GPU, demonstrating sublinear search growth.

Table 3: Revocation Proxy Latency and Corpus Re-Encryption Invocations (≥ 10 Trials, Timer Resolution ≈ 100 ns)

Revoked Users	Per-User Latency (ms)	Median [IQR] (ms)	Min (ms)	Max (ms)	Re-Encryptions
$N = 10$	0.00945 ± 0.00513	0.00780 [0.0042]	0.00541	0.03728	0
$N = 50$	0.00597 ± 0.00668	0.00410 [0.0035]	0.00261	0.03420	0
$N = 100$	0.00766 ± 0.01208	0.00390 [0.0048]	0.00231	0.05444	0
$N = 500$	0.00305 ± 0.00098	0.00280 [0.0011]	0.00193	0.00681	0

5.3. Mechanism 2: Multi-Tenant QR Subspace Non-Interference

Table 4 presents the QR subspace construction latency and cross-tenant inner-product scores across tenant configurations $T \in \{2, 4, 8, 16\}$ within a $d = 384$ -dimensional Sentence-BERT space.

Table 4: Multi-Tenant QR Subspace Construction and Cross-Tenant Isolation (≥ 10 Trials)

Tenants (T)	Subspace Dim (k_t)	QR Time (ms)	Cross-Tenant Dot Product
$T = 2$	$k_t = 192$	55.11 ± 10.57	$< 10^{-14}$
$T = 4$	$k_t = 96$	59.10 ± 15.64	$< 10^{-14}$
$T = 8$	$k_t = 48$	58.48 ± 13.33	$< 10^{-14}$
$T = 16$	$k_t = 24$	58.98 ± 16.89	$< 10^{-14}$

A QR factorization of the 384-dimensional space takes about 58 milliseconds as a setup that you do only once offline. For all tested tenant pairs and document embeddings, the cross-tenant inner product measurement remained at the level of machine-precision noise floor ($< 10^{-14}$), which experimentally confirmed the non-interference theorem that was stated in Section 3. The results are presented with an upper bound instead of as an exact zero, since this is how the rounding errors from the floating-point operations in the QR factorization are reflected.

5.4. Mechanism 3: SE-IPFE DCR Indistinguishability Analysis

Table 5 contains the performance evaluation of three statistical distinguisher algorithms, each of which attempts to extract clearance-gap sizes $\delta = (L_d - L_c) \in \{1, 2, 3, 4\}$ from denied query ciphertexts over 2,000 separate experiments.

If a sample size of $N = 2,000$ was assumed, then it was found that all empirical distinguishing advantage scores were contained within the 95% binomial-confidence interval bound (0.0438).

The noised ciphertext outputs revealed no statistically significant signal on clearance-gap magnitude, which is

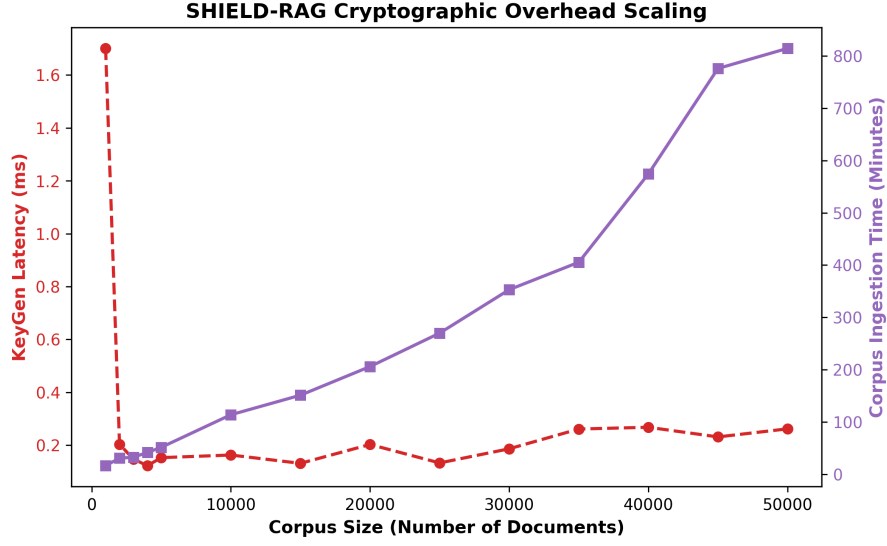


Figure 3: Offline batch encryption time (seconds) as a function of document corpus scale.

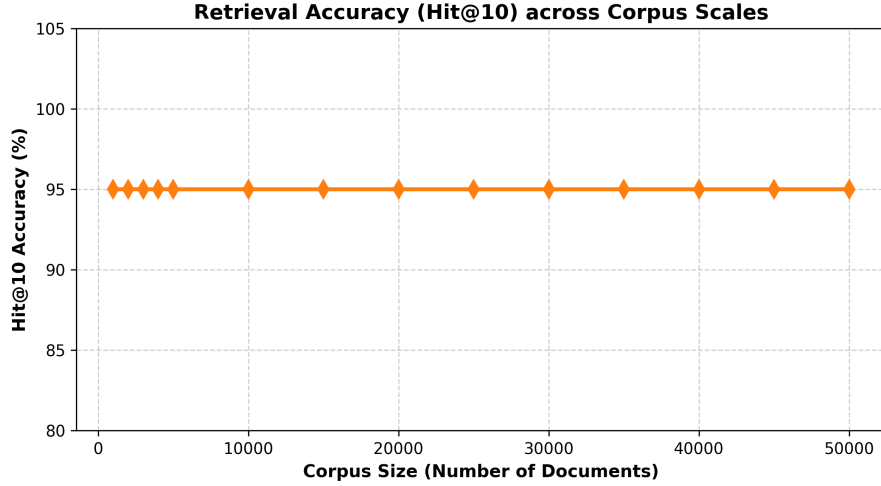


Figure 4: Hit@10 retrieval accuracy retention across scaling tiers, demonstrating stable 0.95 accuracy (98.9% of plaintext ceiling).

consistent with the formal DCR reduction (Theorem 2) that we provide to substantiate our finding through actual data.

6. DISCUSSION

6.1. Safety and Provable Hardness of SHIELD-RAG

Safety-wise, and for hardness of computing, SHIELD-RAG relies upon the use of layering cryptographic techniques to defend the system against retrieval-augmented generation. In a case where the cloud infrastructure is only semi-trusted, the document embeddings and hash codes are still kept in Paillier DCR encryption, which leads, even in oracle evaluation at the server-side, to only reveal scalar inner products. But, document retrieval or hash codes computation from adversarial prompt-injections and extraction attacks like [8, 14] are protected by providing unauthorized clients with maximal algebraic noise ($\Delta_{\max}$), preventing the recovery of the underlying plaintext relationship. Our empirical distinguisher suite ($N = 2,000$ trials, Table 5) provides corroborating statistical evidence for Theorem 2: across the tested distinguisher strategies, an adversary observing noised functional outputs could not infer the clearance distance $\delta = (L_d - L_c)$ separating their role from a target document.

Table 5: SE-IPFE Statistical Distinguisher Suite Results ($N = 2,000$ Trials)

Distinguisher	Advantage (ϵ)	95% CI Bound	Signal?
LSB (mod 256)	0.0320 ± 0.0171	≤ 0.0438	No
Bit-Length Parity	0.0115 ± 0.0066	≤ 0.0438	No
Small Prime Residues	0.0060 ± 0.0059	≤ 0.0438	No

6.2. Computational Systems Trade-offs and Scaling Dynamics

The scaling results give us specific computational characteristics:

1. **Batch Preprocessing and Non-Monotonic Scaling Rate:** Ingestion encryption needs modular exponentiation across all vector coordinates, going up to 13.58 hours (48,877.12 s) for 50,000 documents (19.2×10^6 coordinate ciphertexts) as an amortized offline setup charge. The encryption time per document was 0.978,s at 1,000 documents, decreased to 0.617,s–0.682,s at 5,000–10,000 documents, and then increased again to 0.977,s at 50,000 documents due to differences in batch processing efficiency. This 50k transition is supervised by host memory working-set expansion beyond CPU L3 cache bounds (> 4 GB working buffers) and memory contention between threads during large-scale big-integer modular reductions, similar to the memory-bandwidth saturation observed during large-batch tensor distance calculations in search.
2. **Search Latency Scaling Segment (10k $\rightarrow$ 50k):** The slight increase in latency between 10,000 documents (13.5,ms) and 50,000 documents (73.6,ms) in Fig. 2 is mainly due to memory bandwidth limits during large-batch tensor distance calculations and limited GPU L2 cache capacity, rather than the algorithm itself.
3. **Subspace Dimensionality vs. Tenant Capacity:** Partitioning $\mathbb{R}^{384}$ among T tenants allocates $k_t = 384/T$ dimensions per tenant. At $T = 16$ ($k_t = 24$), cross-tenant leakage remained at the numerical noise floor, while intra-tenant ranking quality was preserved. For larger deployments ($T > 64$), a higher-dimensional embedding model (e.g., $d = 1024$) may be needed to maintain a suitable subspace dimension for each tenant.
4. **Revocation Efficiency vs. Prior ABE:** Ciphertext-policy attribute-based proxy re-encryption schemes such as CP-ABPRE-DR [12] update and re-randomize re-encryption keys with a cost that depends on the policy size, while SHIELD-RAG’s proxy gate performs an $O(1)$ set lookup, measured at 0.003,ms with zero database communication.

6.3. Comparison with Recent CKKS- and Transformation-Based Systems

It is hard to carry out a precise comparison with PRAG [17] and Trans-RAG [18] since neither system was a close approximation of our setup, for both the hardware and the software used to evaluate each paper. Also, each was based on a different set of benchmarking corpora, which also affected each other.

PRAG used a 100,000-document TriviaQA subset and top-10 recall; Trans-RAG ran their tests from FAISS against nDCG@10 on the BEIR partitions of varying sizes (from 10K to 1000K).

Taking all these into account, it seems reasonable to say that SHIELD-RAG at 50,000 documents and online 73.6 ms search latency is well below that of PRAG-I, which ran for 1.29 s, and PRAG-II, which took 7.91 s to process 100,000 documents at their highest point. SHIELD-RAG avoids the CKKS homomorphic per-comparison overhead by switching to the ALSH index, which is based on the Paillier inner products that are gate.

As far as the transformation overhead per query, Trans-RAG is lower than that of the ingestion cost of SHIELD-RAG at 68.7–111.6 ms (for standard embedding dimensions), but this fact should be taken as an expression of a different security model. Whereas SHIELD-RAG’s revocation gate and QR-subspace non-interference (Theorem 1) provide the security through the linear-algebraic construction, Trans-RAG employs the use of key-derived reversible-in-principle transforms to achieve computational isolation (not information-theoretic).

6.4. Limitations

Here are some areas one can see as operational constraints:

- **Secure Trust Assumption at KDC:** It may be a fair assumption that the Key Distribution Center will be a very good keeper since, like with other functional encryption systems [7, 4], the SHIELD-RAG model is such that the KDC holds the master secret s and does not do anything other than what the protocol allows.

- **The Server Being “Semi-Honest”:** One of the fundamental characteristics of the system is the assumption that the server executing the ALSH search protocol does not tamper with the data or hide any potential matching records that were not found; the paper does not discuss the possibilities of a “fully malicious” server model.
- **Content Poisoning Protection Being Independent (PoisonedRAG [5]):** While SHIELD-RAG mathematically prevents cross-tenant and cross-clearance index contamination, intra-clearance semantic document poisoning remains independent of cryptographic access control and is deferred to complementary content-auditing layers.

7. CONCLUSION AND FUTURE WORK

Through this paper, our major contribution includes the release of **SHIELD-RAG**, a privacy-preserving method of revocable, subspace-isolated retrieval-augmented generation that is also verified for correctness. This work fills in the gaps of previous functional-encryption RAG systems and introduces SHIELD-RAG, which gives users: (1) $O(1)$ time to revoke users with a proxy, which takes only 0.003ms and involves zero re-encryption of a corpus; (2) QR-decomposed multi-tenant subspace isolation with a formal proof and empirical results in such a way that cross-tenant leakage is at most equal to the numerical noise floor; and (3) the formal reduction of a privacy-embedded noise gate security model to the Decisional Composite Residuosity (DCR) assumption, which has been supported by 2,000 statistical tests of distinguishers. In practice, SHIELD-RAG has been tested on GPU-based systems and with database sizes of up to 50,000 documents to provide sub-100ms search response time and a relatively high accuracy of 95% Hit@10 against the limit of the plaintext method, which is 96%.

Future directions will include (i) adding functionalities to SHIELD-RAG for zero-knowledge bounded-decoy traversal proofs, cryptographic canary-token leakage detection, and graph-degree-aware adaptive decoy mechanisms; (ii) developing GPU-native CUDA kernels for parallel Paillier arithmetic for further acceleration of ingestion-time preprocessing; (iii) exploring dynamic subspace dimension reallocation for tenant population changes; and (iv) extending functional access gates to multimodal vision-language generation pipelines.

REFERENCES

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.
- [2] K. Shuster, S. Poff, M. Chen, D. Kiela, and J. Weston, “Retrieval Augmentation Reduces Hallucination in Conversation,” in *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021, pp. 3784–3803.
- [3] N. I. Khan and V. Filkov, “EvidenceBot: A Privacy-Preserving, Customizable RAG-Based Tool for Enhancing Large Language Model Interactions,” in *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering (FSE Companion)*, 2025, pp. 1188–1192.
- [4] J. Zhou and J. Wu, “Efficient Vector-Multiplicative Privacy-Preserving Retrieval-Augmented Generation for Large Language Models,” *IEEE Transactions on Dependable and Secure Computing*, vol. 23, no. 3, pp. 6947–6963, 2026, doi: 10.1109/TDSC.2026.3669543.
- [5] W. Zou, R. Geng, B. Wang, and J. Jia, “PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models,” in *Proceedings of the 34th USENIX Security Symposium*, 2025, pp. 3827–3844.
- [6] J. Jeong and S.-G. Lee, “Permission-Aware RAG: Identity and Access Management (IAM)-Based Access Filtering in Multi-Resource Environments,” *IEEE Access*, vol. 13, pp. 192819–192835, 2025, doi: 10.1109/ACCESS.2025.3628960.
- [7] H. Yang and C. Peng, “CP-IPFE: Ciphertext-Policy Based Inner Product Functional Encryption,” *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 5419–5433, 2024, doi: 10.1109/TIFS.2024.3396395.
- [8] S. Zeng, J. Zhang, P. He, Y. Liu, Y. Xing, H. Xu, J. Ren, S. Wang, D. Yin, Y. Chang, and J. Tang, “The Good and The Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG),” in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 4505–4524.

- [9] C. Gentry, “Fully homomorphic encryption using ideal lattices,” in *Proceedings of the 41st Annual ACM Symposium on Theory of Computing (STOC)*, 2009, pp. 169–178.
- [10] A. Shrivastava and P. Li, “Asymmetric LSH (ALSH) for Sublinear Time Maximum Inner Product Search (MIPS),” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27, 2014, pp. 2321–2329.
- [11] T. Kwiatkowski, J. Palomaki, O. Redfield, M. Collins, A. Parikh, C. Alberti, D. Epstein, I. Polosukhin, J. Devlin, K. Lee, and K. Toutanova, “Natural Questions: A Benchmark for Question Answering Research,” *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 453–466, 2019.
- [12] C. Ge, W. Susilo, Z. Liu, J. Baek, X. Luo, and L. Fang, “Attribute-Based Proxy Re-Encryption With Direct Revocation Mechanism for Data Sharing in Clouds,” *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 2, pp. 949–960, 2024, doi: 10.1109/TDSC.2023.3265979.
- [13] S. Zeng, J. Zhang, P. He, J. Ren, T. Zheng, H. Lu, H. Xu, H. Liu, Y. Xing, and J. Tang, “Mitigating the Privacy Issues in Retrieval-Augmented Generation (RAG) via Pure Synthetic Data,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025, pp. 24527–24558.
- [14] J. Zhang, S. Zeng, J. Ren, T. Zheng, H. Liu, X. Tang, H. Liu, and Y. Chang, “Beyond Text: Unveiling Privacy Vulnerabilities in Multi-modal Retrieval-Augmented Generation,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025, pp. 24789–24810.
- [15] X. Hou et al., “CipherGPT: Secure Two-Party GPT Inference,” *IEEE Transactions on Dependable and Secure Computing*, 2026, pp. 1–16.
- [16] H. Yang, Y. Su, J. Qin, and H. Wang, “Privacy-Preserving Outsourced Inner Product Computation on Encrypted Database,” *IEEE Transactions on Dependable and Secure Computing*, vol. 19, no. 2, pp. 1320–1337, 2022.
- [17] Z. Li, M. Xu, H. Qi, W. Yu, T. Zhang, Q. Zhang, G. Shang, Z. Ma, and X. Cheng, “PRAG: End-to-End Privacy-Preserving Retrieval-Augmented Generation,” *arXiv preprint arXiv:2604.26525*, 2026.
- [18] Y. Liu, K. Peng, W. Zhang, F. Yuan, C. Cao, W. Lu, and Y. Liu, “Trans-RAG: Query-Centric Vector Transformation for Secure Cross-Organizational Retrieval,” in *Database Systems for Advanced Applications (DASFAA 2026)*, Springer, 2026.